

- Incubator stage was built in water reservoir to maintain humidity

- Uses Continuous Current Control instead of PWM with LX-W lid

- One lid and water-reservoir units can support: multi-well → ATX-W dish

with
• LX-D35 lid
• LX-D56 lid

← ATX-D dish

- 35mm dish
- 60mm dish

with
LX-CSG lid

← ATX-CSG dish

- slide
- chamber slide
- chamber cassette

Note Stage

WSKM X Series

Supports Zeiss K type Pump

- Heater → STXG built-in digital Gas mixer
- STXF built-in analogue flow meter

STX - Series (Stage Top Series)

Controller:

STXG/STXF → model Temperature Controller & CO₂ Controller

Chamber:

WSKM → model for chamber designed to fit Zeiss K type frame

↳ Includes: 1. TOP heater → heated glass lid

2. Bath unit → mounting point for dishes + provides chamber humidification

3. Stage heater → Heats dish attachment

↳ supports several attachments for different sample holders

Dish attachment:

• ATX-D: 35/50/60 mm dish (uses LX-D35 / LX-D56 lids) Attaches to ATX-A

• ATX-CSG: slide glass / chamber slide / coverglass cs (uses LX-CSG lid) Attaches to ATX-A

- **ATX-w** : WellPlate (uses LX-w lid) directly attaches to chamber
- **ATX-A** : Mounting plate for ATX-D/ATX-CSG Attaches to ATX-w

Add-ons

- do not let air flow directly over the chamber to prevent focus drift during long-term observations
- avoid intense vibration or impact
- possible problems with environment with airflow!

SPECS :

STXG : • Built-in Gas mixer (STXF model has an analog flow valve with fixed 5% CO₂ mix)

(Temp / CO₂

controller)

(STXF In block)

• Dimensions (W x D x H mm) : 160 x 280 x 256

• Weight : 5.4 Kg

• Temp controlled by PID loop with continuous current control
Temperature

• Temp Ranges : → Top Heater : Ambient + 5°C — 65°C
(min 10°C)

↳ Bath Heater : Ambient + 5°C — 50°C

↳ Stage Heater : Ambient + 5°C — 50°C

↳ Lense Heater : Ambient + 5°C — 45°C

• Temp Accuracy : $\pm 0.3^{\circ}\text{C}$ at heater surface (in specific conditions)

• Temp Resolution : 0.1°C (digital)

• Temp Rise time : 10 mins to get to 50°C

• Temp Sensor : Pt 100

Humidity

• Humidity Forced by heated water bath in chamber

Gas control

• O₂ Range : 5% — 20% (STXF model fixed to 5%)

• CO₂ Accuracy : $\pm 0.1\%$

• Gas Input : 100% CO₂ @ 1 — 1.5 Bar (STXF model requires compressed Air with 5% CO₂ pre mix)

• Gas output : 160 ml/min (STXF model 75 — 250 ml/min)

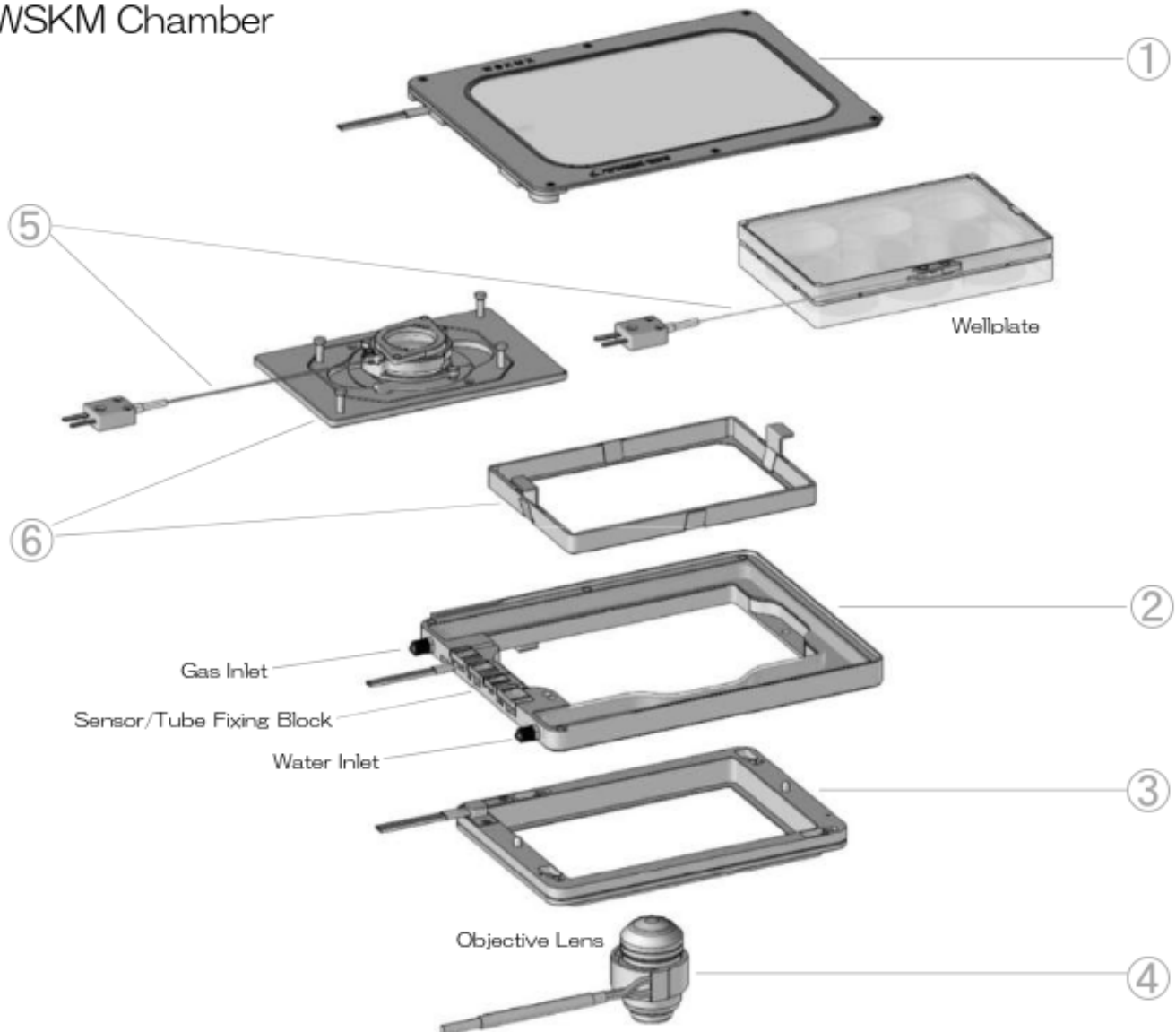
Input : Input Power 110W

Use conditions : Indoor

Temp : 20°C — 28°C

RH: 35% - 85%
 Altitude: up to 2000 m max
 Environmental: Inst cat of IEC 6064, Pollution degree 2

WSKM Chamber



No.	Name	Role	Note
①	Top Heater	• Condensation prevention • Sample temperature control	
②	Bath Unit	• Humidification inside of chamber by steam	Presence/absence of Bath Heater depends on the chamber model.
③	Stage Heater	• Heats Dish Attachment • Prevents heat from escaping to the stage.	
④	Lens Heater	• Prevents heat from escaping the objective lens.	For use of Lens Heater, see P.24.
⑤	Feedback Sensor	• Measurement of sample temp.	
⑥	Dish Attachment & Fixing Lid	• Fixation of vessel	Dish Attachment and Fixing Lid differ depending on the chamber model.

• WSKM X . Dimensions W187 X D146 X H127 mm

• Recommended 40°C water in heated Bath

Can System work without feedback sensor?

• Feedback sensor is set to dip directly into the media!

↳ Feedback sensor is used for measurement of Sample temp

↳ Feedback control is used by feedback control loop to adjust the Top Heater temperature to better regulate sample temp.

otherwise the stage / Bath temp are set to the temp required by sample, and top is slightly higher to prevent condensation
other heated parts have integrated sensors so as like

• 3 access ports → port for 2 tubes (Silicone < 3mm) for perfusion
↳ Sensor port for feedback sensor
↳ Port for 1 tube (Silicone < 3mm) for drug administration
→ can set each port to open or closed

• Optional Gas modules can be utilized for additional functionality

→ STX-CO202 → takes over CO₂ control
↳ Provides O₂ control

STX6 only supports STX-CO202

STX6 supports both STX-CO202

and STX-CO2
+ STX-O2

• General work-flow is to start preliminary operation about an hour before experiments to let temp stabilize (using dummy plate)

→ Then 30 mins before experiment, start Gas supply to stabilise Gas

• Generally set the heater temps slightly higher than desired sample temp to allow for temp gradient

• Computer software can change and log env settings

Conditions:

- Manual indicates to not operate in environment with intense vibration or impact. What is intense?, how much can it withstand
- Manual indicates that room temp should be kept stable between 20-28°C and to not let air flow directly over the chamber.
Manual indicates that this is to avoid focus drift during observations.
Do these restrictions apply only to optical conditions or is it also for Incubation chamber conditions?
- In FXQ, they recommend direct impact from air-conditioner to stabilize the heaters